

3 (a) State what is meant by the *internal energy* of a gas.

.....

.....

..... [2]

(b) The volume occupied by 1.00 mol of liquid water at 100°C is $1.87 \times 10^{-5} \text{ m}^3$. When the water is vaporised at an atmospheric pressure of $1.03 \times 10^5 \text{ Pa}$, the water vapour has a volume of $2.96 \times 10^{-2} \text{ m}^3$.

The latent heat required to vaporise 1.00 mol of water at 100°C and $1.03 \times 10^5 \text{ Pa}$ is $4.05 \times 10^4 \text{ J}$.

Determine, for this change of state

(i) the work done on the system,

work done on the system =J [2]

(ii) the heat supplied to the system,

heat supplied to the system =J [1]

- (iii) the change in internal energy of the system. State whether the change is an increase or a decrease in internal energy.

change in internal energy = J
..... [2]

- (c) Using your answer to (b)(iii), estimate the binding energy per molecule in liquid water.

binding energy per molecule =J [2]

[Total: 9]